Eralp Çelebi

Sakarya, Türkiye | eralp.celebi.personal@gmail.com | github.com/EralpCelebi | linkedin.com/in/eralp-çelebi-5bb731194/

Education

Sakarya University — Sakarya, Türkiye
Electrical-Electronics BEng. **(2.96/4.00)** — September 2022 • June 2027

Tomaš Batá University — Zlín, Czechia
Erasmus+ Exchange Student — February 2025 • June 2025

Experience

GESK Technologies | Embedded Systems Engineering Intern — August 2024
C, Makefile, Yocto, Linux, ARMv8 — *Sakarya, Türkiye*

- Developed and deployed custom embedded operating systems tailored specifically for the Raspberry Pi 3 B+ hardware platform by utilizing the Yocto Project and GNU Make to build optimized Linux distributions.
- Enabled low-level communication between the OS and proprietary hardware through the creation and integration of custom Linux kernel modules by developing character drivers in C to interface with custom hardware cards via I2C.

Academic Projects

TÜBİTAK • 2209-B — 2025 • 2026
C, Go, ESP32, SX1276 — (tubitak.gov.tr/tr)

- Designed an ESP32-based distributed control and telemetry system to optimize data collection processes in industrial factory environments.
- Developed a communication gateway utilizing LoRa for long-range transmission and I2C for interfacing with local hardware sensors.
- Implemented a wireless network topology to reduce cabling costs and enable real-time monitoring across large-scale industrial facilities.

Projects

OaSYS • Operating System — 2024 • 2026
C, CMake, UEFI, ARMv8, x86_64 — (github.com/EralpCelebi/oasys)

- Developed a Unix-like operating system kernel from scratch in C to manage process scheduling and hardware abstraction layers.
- Enhanced system initialization efficiency across diverse hardware configurations by implementing a modern build system and UEFI support.

Damascus • Boot Loader — 2024
x86_64 Assembly (NASM), GNU Make — (github.com/EralpCelebi/damascus-boot)

- Architected an x86_64 bootloader in pure Assembly to initialize system hardware and manage CPU transition into protected mode.
- Optimized memory addressing routines to ensure reliable loading of kernel sectors into RAM within BIOS constraints.

Skills

- **Programming Languages:** C, Rust, Go, Assembly (x86_64, AArch64), Python, Julia, MATLAB
- **Design Software:** KiCAD, SPICE
- **Software:** Git, Docker, GNU Make, CMake, LLVM, GDB